

Lecture 20

Generalised hydrodynamics, and the quantum Newton's cradle

Objectives. Course map: P4 and P5, the second half of G4, and the course closed. [Lecture 19](#) proved, for one field quench of the Ising chain, that the transverse magnetisation relaxes to its value in a generalised Gibbs ensemble. In an interacting gas the ensemble is one function, the filling fraction of [Lecture 17](#); let it vary slowly in space and it acquires a dynamics, generalised hydrodynamics, at the dressed velocity. Hard rods show what the dressed velocity is, and the trap of [Lecture 0](#)'s Newton's cradle shows why its clouds kept returning. The lecture ends by closing [Lecture 0](#)'s map.

20.1 The question

[Lecture 0](#) began with two clouds that passed through each other thousands of times and did not become one thermal hump. [Lecture 17](#) gave the gas its equation of state, and [Lecture 19](#) proved relaxation for a quench of the Ising chain; here the gas is assumed to be locally in a generalised Gibbs ensemble. So the question is

how does an integrable gas flow when its state varies in space, and what did the cradle show?

The answer: the density of each rapidity is carried at its dressed velocity ((20.5), (20.4)); for hard rods the dressed velocity is computed in one line ([Proposition 20.1](#)), and in the trap's two free limits the flow is a rigid rotation that returns exactly ([Figure 20.3](#)). The worked example is hard rods.

20.2 Interacting models: one function carries the state

Free fermions made the argument exact, and easy, because the charges were quadratic. The replacement for an interacting model is already built: a macrostate in the thermodynamic limit is a root density $\rho(\lambda)$, or the filling fraction $\vartheta = \rho/\rho_t$ of [Definition 17.1](#), and every extensive charge is

$$\frac{\langle Q_a \rangle}{L} = \int q_a(\lambda) \rho(\lambda) d\lambda, \quad (20.1)$$

with q_a the charge's one-particle eigenvalue — $q = k^2$ for the energy of the gas, $q = k$ for its momentum. A family of charges fixes ρ exactly when its q_a separate root densities; given such a *complete* family, the generalised Gibbs ensemble of [Definition 19.5](#) is a *single function* — for a one-species gas; one filling per quasiparticle species in general, as the chain's strings ([Lecture 17](#)'s reading) already needed — the same ϑ that [Lecture 17](#) minimised the free

energy over, now fixed by the initial state instead of by a temperature. Maximising (17.4) at fixed (20.1) is the generalised thermodynamic Bethe ansatz, and it returns (17.7) in dimensionless form: with $\eta = \ln(\rho_h/\rho)$ and $w = \sum_a \beta_a q_a$,

$$\eta(\lambda) = w(\lambda) - \frac{1}{2\pi} \int K(\lambda - \mu) \ln(1 + e^{-\eta(\mu)}) d\mu, \quad \vartheta = \frac{1}{1 + e^\eta} \quad (20.2)$$

· *cited*. The multipliers carry the temperature, so (17.7)'s explicit T 's go: at thermal equilibrium $\eta = \varepsilon/T$, and $w = (\varepsilon_0 - \mu)/T$ returns Lecture 17's equation exactly.

20.3 Generalised hydrodynamics

Now let the state vary in space: $\vartheta(\lambda, x, t)$, slowly on the scale of the interactions — a *local* generalised Gibbs ensemble. That is the assumption, and the only one; there is no gradient expansion beyond leading order, which is what *Euler scale* means.

How fast does a quasiparticle actually move?

Lecture 12 found that a collision permutes the momenta and leaves a *displacement* behind (Theorem 12.2), and Lecture 11 found its quantum counterpart (11.7). A quasiparticle crossing a gas therefore collects one displacement per particle it passes, and does not travel at its bare velocity.

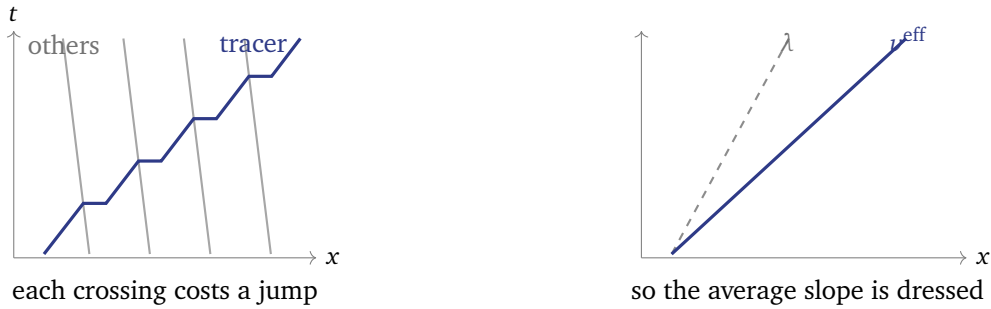


Figure 20.1: The effective velocity. *Left*: a tracer (dark) crossing a gas; each crossing displaces it, here forwards. *Right*: over many crossings the bare slope λ becomes an average slope v^{eff} , which depends on the whole state through the rate at which crossings happen. For hard rods the bookkeeping closes in one line, (20.3); in general into Lecture 17's dressing equation, (20.4).

Proposition 20.1 (Hard rods · *sketched*). Let the gas be N identical rods of length a and unit mass on a line, moving freely between elastic collisions, so that the bare momentum is $p_0(\lambda) = \lambda$ and the bare energy $\varepsilon_0(\lambda) = \lambda^2/2$. Equal masses exchange velocities, so the set of velocities is only permuted and the gas is integrable; its rapidities are the velocities. A tracer of velocity λ jumps forward by a each time it passes a rod, so

$$v^{\text{eff}}(\lambda) = \lambda + a \int d\alpha \rho(\alpha) [v^{\text{eff}}(\lambda) - v^{\text{eff}}(\alpha)] = \frac{\lambda - aJ}{1 - an}, \quad n = \int \rho, \quad J = \int \alpha \rho. \quad (20.3)$$

Proof. The rate at which the tracer passes rods of velocity α is $\rho(\alpha) |v^{\text{eff}}(\lambda) - v^{\text{eff}}(\alpha)|$ and the jump is $\pm a$ with the sign of the relative velocity, which is the first equality. Integrating it against ρ gives $\int \rho v^{\text{eff}} = J$ identically, and substituting that back solves it. ■

Equation (20.3) is Equation (17.1) with the *constant* kernel $K = -a$ — not a local one: a delta-function kernel would dress every function by the same factor and cancel in the ratio. Dressing with it gives $2\pi\rho_t = (1)^{\text{dr}} = 1 - an$, the free volume fraction, and $(\varepsilon'_0)^{\text{dr}} = \lambda - aJ$, whose ratio is (20.3). Lecture 17's large- c limit (the paragraph after Proposition 17.9) says this is no curiosity: to leading order in $1/c$, and in the rest frame, the Lieb–Liniger gas is free fermions in a box *lengthened* by $2/c$ per particle, that is, a hard-rod gas of length $a = -2/c$. In general

$$v^{\text{eff}}(\lambda, x, t) = \frac{(\varepsilon'_0)^{\text{dr}}}{(k'_0)^{\text{dr}}}, \quad (20.4)$$

with the dressing of (17.1) at the local ϑ and k_0 the excitation's *own* bare momentum (Remark 17.3). Write it this way and not as $\varepsilon'/2\pi\rho_t$, which is Equation (17.10). Lecture 17 got that by differentiating a *thermal* Yang–Yang equation driven by $\varepsilon_0 - \mu$; differentiating (20.2) gives $\eta' = (w')^{\text{dr}}$ instead, which dresses the *combination* w and not the physical energy. (Free fermions with $w = \lambda^4$: there $2\pi\rho_t = 1$ and $\eta' = 4\lambda^3$, against 2λ from $\varepsilon_0 = \lambda^2$.) The two coincide at thermal equilibrium, where $\eta = \varepsilon/T$ and the T cancels from the ratio — which is where Lecture 17 used it, saying explicitly that calling the ratio a propagation speed was not proved there. This section is where it is used, and the rods are the reason to believe it.

The equation

$$\partial_t \rho(\lambda, x, t) + \partial_x [v^{\text{eff}}(\lambda, x, t) \rho(\lambda, x, t)] = 0 \quad \text{for every } \lambda. \quad (20.5)$$

One continuity equation per rapidity · *sketched: Castro-Alvaredo, Doyon and Yoshimura 2016; Bertini, Collura, De Nardis and Fagotti 2016*: the number of quasiparticles of each rapidity is separately conserved — what integrability buys — and they are transported at the dressed velocity, which is what interaction costs. Ordinary hydrodynamics has five such equations in three dimensions; this has one per point of a continuum.

Figure 20.2 is the theory in one picture, and the one place in this course where an Euler-scale claim is tested against the microscopic model it describes: the circles are rods, evolved exactly, not a second solution of (20.5). The agreement is a law of large numbers and needs many particles per hydrodynamic cell — at one rod per ten units of length the same comparison fails outright.

20.4 The Newton's cradle, finally

Equation (20.5) has an equivalent form: the filling fraction is not conserved but *advected*, $\partial_t \vartheta + v^{\text{eff}} \partial_x \vartheta = 0$ in the homogeneous case — not obvious, since v^{eff} depends on ϑ · *cited: Castro-Alvaredo, Doyon and Yoshimura 2016*. There a trapping potential $V(x)$ enters as a force on the rapidity, and the content is that quasiparticles follow characteristics:

$$\partial_t \vartheta + v^{\text{eff}} \partial_x \vartheta - \partial_x V \partial_\lambda \vartheta = 0, \quad \text{i.e.} \quad \dot{x} = v^{\text{eff}}, \quad \dot{\lambda} = -\partial_x V, \quad (20.6)$$

with ϑ constant along them · *cited: Doyon and Yoshimura 2017*. In Lecture 11's units, $\hbar = 2m = 1$, the rapidity is the momentum: λ obeys Newton's second law and x a dressed version of the first.

One caution first: (20.6) does not say the trapped model is integrable. A position-dependent potential breaks the homogeneous charges, no Bethe ansatz solves the trapped

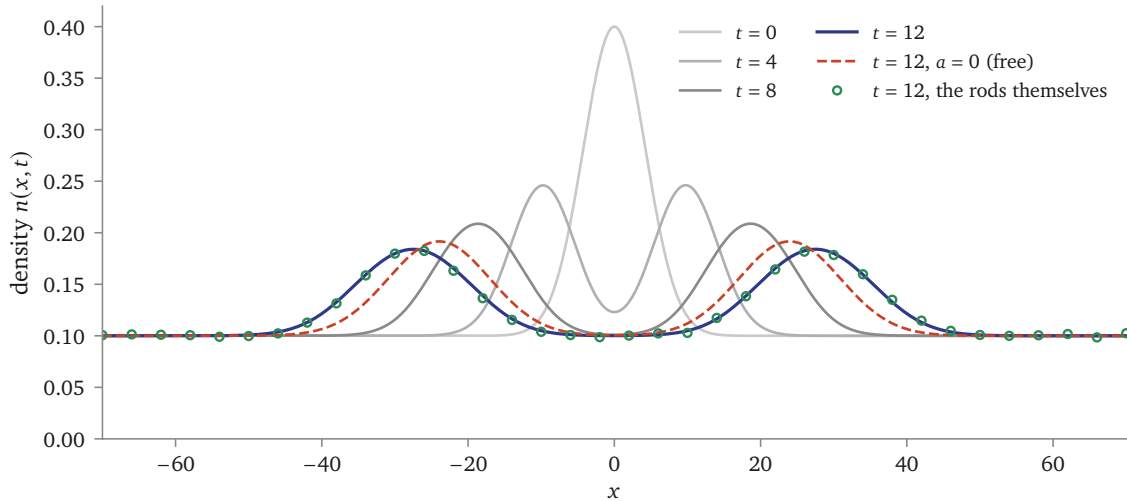


Figure 20.2: A lump of hard rods splitting, from (20.5): an excess of rods on a uniform background (palest) with rapidities near $\pm\lambda_0$ separates into a left-mover and a right-mover — the Newton's cradle with one collision instead of thousands. The rods have $a = 1$ and an reaches 0.40, so the dressing is large: the dashed curve is the *same* initial state evolved with $a = 0$, its peak 3.55 units behind. The circles are a direct simulation of the rods — no hydrodynamics — prepared as a genuine non-overlapping configuration and lying on the GHD curve to 3% of the lump height. The check script verifies that the solver conserves the number of rods to 10^{-12} , separately in every rapidity slice.

Lieb–Liniger Hamiltonian, and (20.6) is an Euler-scale approximation, good while V varies slowly on the scale of the interactions · *cited: Doyon and Yoshimura 2017; Caux et al. 2019, App. H*. It says two things.

First, at the two ends of the coupling range the trap costs nothing at all. There $\nu^{\text{eff}} = 2\lambda$ — the ideal gas, and the impenetrable limit, where Girardeau's map makes the gas free fermions in every state (Lecture 18's reading) — and (20.6) is a *linear* flow of period $2\pi/\omega$: the distribution turns rigidly on ellipses of fixed $x^2 + (2\lambda/\omega)^2$ and returns exactly to itself (Figure 20.3). That is the cradle. In between, ν^{eff} depends on the local state, the return is only approximate and the shape drifts — which is what the experiments see. Second, $\int \rho \, dx$ is in general *not* conserved in a trap, since (20.6) moves λ : “the momentum distribution does not relax” means a shape returning, not a constant.

The theory, and then the data. Caux, Doyon, Dubail, Konik and Yoshimura applied GHD to the cradle geometry in 2019, at a coupling close to the original experiment's. The first experimental test is Schemmer, Bouchoule, Doyon and Dubail's atom chip: one cloud after a longitudinal trap quench, where GHD works and conventional hydrodynamics, which assumes local *thermal* equilibrium, does not. The quantitative test is Malvania, Zhang, Le, Dubail, Rigol and Weiss, in the same laboratory's gases, measuring rapidity distributions directly through dozens of trap oscillations at $\gamma = c/n$ from 0.3 to 9.3 · *cited*. **There is, as of writing, no published comparison of generalised hydrodynamics with the 2006 data itself;** theory and experiment meet in the later apparatus, not in the original plot.

What else spoils it. The atoms of Lecture 0's tube are not exactly one-dimensional: virtually exciting a transverse mode and coming back leaves an effective *three*-body collision along the tube, which no Bethe ansatz solves, and whose rate sets a thermalisation time · *cited*:

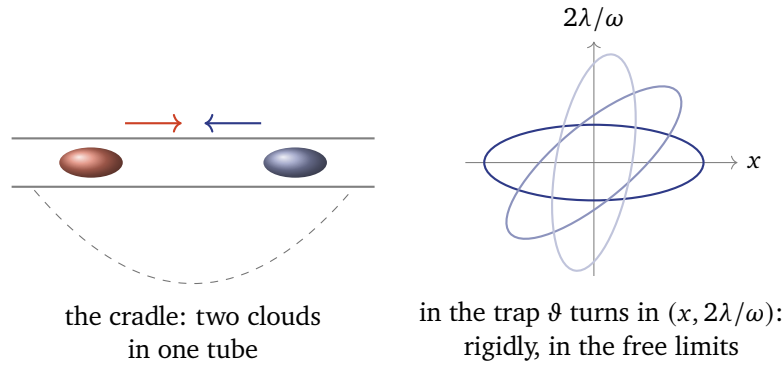


Figure 20.3: [Lecture 0's cradle](#) ([Figure 2](#)), and what generalised hydrodynamics says about it. A harmonic trap adds the force term of (20.6). In the two free limits the characteristics are linear: the distribution turns rigidly in the scaled plane $(x, 2\lambda/\omega)$ at the trap frequency and returns to itself each period, so the clouds pass through each other and re-form, thousands of times. At intermediate coupling ν^{eff} depends on the local state, the turning is not rigid and the return is only approximate. The free skeleton of the mechanism, not a claim about every coupling.

Mazets, Schumm and Schmiedmayer 2008. Integrability is therefore *approximate* on two counts, and the question is one of time scales.

Remark 20.2 (Prethermalisation, and the quantum KAM question). Under weak integrability breaking a system started far from equilibrium first relaxes, on the fast scale, to the generalised ensemble of the *unbroken* model — a *prethermal* plateau — and only much later drifts to the thermal state, on a scale that grows as the breaking is reduced · *cited: Bertini, Essler, Groha and Robinson 2015; observed in Gring et al. 2012.* This is the nearest quantum relative of [Theorem 7.8](#), and the analogy is honest about where it stops: KAM is a theorem, with hypotheses, about which tori survive; prethermalisation is a statement about time scales with no theorem behind it. Whether approximate charges survive weak breaking in a controlled sense is the *quantum KAM* question, and it is open · *cited: Brandino, Caux and Konik 2015.*

20.5 The map, closed

[Lecture 0's](#) map named five ideas. Here is what each half of the course proved about them, and what it cited.

- *P1, integrability as a count.* Part I proved it, with the statuses its lectures give: the Liouville–Arnold theorem ([Theorem 5.7](#)), action–angle variables ([Theorem 6.5](#)), and the Toda ring integrable ([Theorem 9.10](#)). In Part II the count is not a definition ([Lecture 0's](#) caveat); a commuting family from one matrix takes its place.
- *P2, one family from one local identity.* Proved on both sides: the quadratic bracket makes the Toda ring's transfer matrices commute ([Corollary 9.8](#)), *RTT* the Heisenberg chain's ([Theorem 14.3](#)), and the quantum Toda chain proves that the second becomes the first as $\hbar \rightarrow 0$ ([Corollary 14.6](#)).
- *P3, scattering that only rearranges momenta.* The classical side cites Moser ([The-](#)

orem 12.2); the quantum side derives the Yang–Baxter equation from three-body consistency (Theorem 13.3) and the Bethe equations from a ring: for the gas with a full proof that the Bethe wavefunctions are eigenstates (Lecture 11), and for the chain sketched in coordinates (Lecture 10) and proved algebraically (Lecture 15).

- *P4, densities, dressing and flow.* Quantum only. The density equation is proved from an assumption (Lecture 16), the Yang–Yang equation from a count (Lecture 17), and the hydrodynamics of this lecture is sketched, with hard rods as its check.
- *P5, relaxation to a generalised ensemble.* Proved for one quench and one observable of the Ising chain (Theorem 19.7); for the interacting gas it is the premise of this lecture's hydrodynamics, and the cradle is only nearly integrable.

The dictionary between the two halves, typed: **T** a theorem of this course, an explicit proved map or $\hbar \rightarrow 0$ limit between the columns; **A** a structural analogy, one role played by two separately established facts; **C** a citation. The column L is the lecture, or the optional lecture, where the Part II side lives.

Part I	Part II	L	type
Poisson bracket $\{\cdot, \cdot\}$	commutator $[\cdot, \cdot] / i\hbar$	14	A
n integrals in involution	no agreed definition; criteria	0, A	A
Poincaré sections	level statistics	A	A
linear action–angle system	free fermions, $\sum \varepsilon(k)n_k$	18	A
$\oint p \cdot dq = 2\pi\hbar n$	Bethe equations, logarithmic form	10	A
permuted momenta, braid	factorised S-matrix, Yang–Baxter	13	A
classical displacement	wave-packet displacement $\mp\theta'$	11	A
Lax matrix $L(\lambda)$	Lax operator $L_n(\lambda)$	14	T
Sklyanin bracket	RTT	14	T
$r(\lambda)$	$R/\lambda = 1 + i\hbar r + O(\hbar^2)$	14	T
$\{t(\lambda), t(\mu)\} = 0$	$[t(\lambda), t(\mu)] = 0$	14	T
$\text{tr } L^k$	$d^k \ln t(\lambda) _0$	15	A
classical Yang–Baxter	Yang–Baxter	14	T
spectral curve $\det(z - T)$	T–Q relation	15	A, C
bands and gaps	Fermi sea of rapidities	16	A
frequency $\partial K / \partial I$	dressed velocity	17	A
measure $dI d\varphi$	Yang–Yang entropy	17	A
CM by reduction; isochrony	CS; equally spaced trapped spectrum	B	A
$\mathfrak{sl}_N$ roots	Dynkin-indexed nested equations	C	A, C
elastic non-diffractive scattering	labels transported by X	13, C	A
invariant tori	generalised Gibbs ensemble	19	A
KAM	prethermalisation	20	A, C

20.6 What it buys

- *One continuity equation per rapidity* (20.5), with the dressed velocity (20.4) as the speed at which each rapidity is carried; hard rods show why (Proposition 20.1).
- *The cradle:* in the two free limits the trapped distribution turns rigidly and returns exactly; between them it returns approximately.
- *The course closed:* what each idea of Lecture 0's map rests on, proved, sketched or cited.

20.7 Caveats, in one place

- *Status.* [Proposition 20.1](#) and (20.5) are sketched; the generalised thermodynamic Bethe ansatz, the advection form and the trap equation (20.6) are cited.
- *Euler scale.* Generalised hydrodynamics is leading order in the gradients; the trap breaks the homogeneous charges, and the tube is not exactly one-dimensional, so the cradle is integrable only approximately.
- *No comparison with the 2006 data* has been published; theory and experiment meet in later apparatus.
- *A complete family of charges* is needed to fix the state, and for the XXZ chain the ultralocal ones are not enough (the reading).
- *Only the rows marked T are theorems*, and all of them are [Lecture 14's](#).

Reading

20.8 When the ultralocal charges are not enough

The ultralocal charges are not enough. For the XXZ chain the charges generated by $t(\lambda)$ as in [Lecture 15](#) do not fix ϑ : two macrostates with different ϑ can share all of them, and quasi-local charges — the ones whose densities decay exponentially instead of stopping, exactly as $Q[\cos mk]$ did in [Lecture 19](#) — are needed to close the ensemble · *cited: Ilievski, De Nardis, Wouters, Caux, Essler and Prosen 2015*. This is where the free-fermion picture misleads if pushed: there the strictly local charges sufficed, because ε never vanished.

Reading

20.9 More than one family

Remark 20.3 (More than one family). The labelled gases of optional lecture C carry two coupled families of rapidities, charge and spin. Nothing above changes in form: the macrostate is a pair $(\rho, \rho_{\text{spin}})$, the charges are integrals against both, and the generalised ensemble is a pair of filling fractions coupled through the nested Bethe equations of optional lecture C. What changes is that the spin family carries no energy of its own ($E = \sum k_j^2$), so the energy does not determine how the spin rapidities are distributed. It does not follow that the energy says nothing about them: the spin roots enter the charge equations and move the charge roots, so fixing the energy can exclude whole spin sectors — at $N = 2$, $L = c = 1$ the zero-momentum singlet has $E = 1.8439$ while every triplet state has $E \geq 4\pi^2$ — and a thermal saddle couples the two families. What a generalised ensemble can do is be told a spin distribution that no temperature would have chosen.

Summary.

- *One function carries the state:* with a complete family of charges, the generalised Gibbs ensemble of a gas is a filling fraction ϑ , fixed by the generalised thermodynamic Bethe ansatz.
- *Hard rods* ([Proposition 20.1](#)): a tracer jumps by a at each rod, and $v^{\text{eff}} = (\lambda - a_j)/(1 - an)$.

- *Generalised hydrodynamics* (20.5): each rapidity's density is carried at v^{eff} , the dressed velocity.
- *The cradle*: in a trap the distribution turns and returns, exactly in the two free limits.
- *Beyond the path*: optional lectures A (quantum chaos and level statistics), B (the Calogero–Moser systems) and C (nesting and the Hubbard model).

Tutorial. Exercise Sheet 20.

References

O. A. Castro-Alvaredo, B. Doyon and T. Yoshimura, *Phys. Rev. X* **6** (2016) 041065, and B. Bertini, M. Collura, J. De Nardis and M. Fagotti, *Phys. Rev. Lett.* **117** (2016) 207201 (generalised hydrodynamics); B. Doyon, *SciPost Phys. Lect. Notes* **18** (2020) (hard rods, Example 3.4; the external force, §4.5) and B. Doyon and T. Yoshimura, *SciPost Phys.* **2** (2017) 014; C. Boldrighini, R. L. Dobrushin and Yu. M. Sukhov, *J. Stat. Phys.* **31** (1983) 577 (hard rods); E. Ilievski *et al.*, *Phys. Rev. Lett.* **115** (2015) 157201 (quasi-local charges); J.-S. Caux, B. Doyon, J. Dubail, R. Konik and T. Yoshimura, *SciPost Phys.* **6** (2019) 070 (the cradle geometry); M. Schemmer, I. Bouchoule, B. Doyon and J. Dubail, *Phys. Rev. Lett.* **122** (2019) 090601, and N. Malvania *et al.*, *Science* **373** (2021) 1129 (the experiments); I. E. Mazets, T. Schumm and J. Schmiedmayer, *Phys. Rev. Lett.* **100** (2008) 210403; B. Bertini, F. H. L. Essler, S. Groha and N. J. Robinson, *Phys. Rev. Lett.* **115** (2015) 180601 (prethermalisation); G. P. Brandino, J.-S. Caux and R. M. Konik, *Phys. Rev. X* **5** (2015) 041043.